Quadratics

Chapter Overview 2

1. Solving quadratic equations
Factorisation, the quadratic formula and substitution.

2. Completing the square
Rewriting and solving quadratic expressions and equations.

3. Quadratics as functions
Inputs, outputs, roots and minimum values.

4. Quadratic graphs
Intercepts, turning points and equations from graphs.

5. The discriminant
How many real roots?

6. Modelling with quadratics
Using and interpreting quadratic models.

Why do we care about quadratics... 3

Quadratic relationships arise in many practical situations.

Summations

$$1 + 2 + \cdots + n = \frac{1}{2}n^2 + \frac{1}{2}n$$

A sports event awards 8 points, then 7, then 6, and so on. For $1 \leq n \leq 8$ competitors:

$$8 + 7 + \cdots + (9 - n) = \frac{n(17 - n)}{2}$$

The totals for one, two and three competitors are 8, 15 and 21.

Projectile motion

Under constant acceleration,

$$s = ut + \frac{1}{2}at^2$$

Here s is displacement, u initial velocity, a acceleration and t time. The expression is quadratic in t.

A projectile moving under gravity alone follows a parabolic path.

Why do we care about quadratics... 4

Hannah's sweets There are n sweets in a bag, of which 6 are orange. Hannah eats two sweets selected at random, without replacement. The probability that both are orange is $1/3$.

Show that $n^2 - n - 90 = 0$.

Hannah's sweets

A product of two expressions involving the same variable often produces a quadratic.

Solve $x^2 + 5x = 6$. Two methods are shown below; completing the square is a third method.

1. Factorisation First put the equation in the form $ax^2 + bx + c = 0$.

1. Factorisation

2. Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2. Quadratic formula

Solving without expanding

$(x - 1)^2 = 5$

Solving without expanding

Quadratics in disguise

In $x - 6\sqrt{x} + 8$, the expression is quadratic in $\sqrt{x}$. Use substitution or recognise the factors directly.

Substitution Solve $x - 6\sqrt{x} + 8 = 0$, using $y = \sqrt{x} \geq 0$.

Substitution

Direct factorisation Solve the same equation without substituting.

Direct factorisation

Test Your Understanding 7

1 Solve $(x+3)^2 = x+5$ by factorisation.

1

2 Solve $(2x+1)^2 = 5$.

2

3 Solve $\sqrt{x+3} = x-3$.

3

4 Solve $2x + \sqrt{x} - 1 = 0$.

4

Calculator check: use polynomial/equation mode with degree 2 for $ax^2 + bx + c = 0$. Show the algebra when the question requires it.

Exercise 2A/2B Pure Y1 Pages 20-22 8

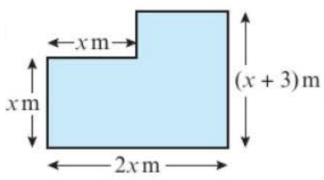
3(i) Solve $6x^2 - 7 = 11x$.

3(i)

3(h) Solve $5x^2 - 10x^2 = -7 + x + x^2$.

3(h)

4 This shape has an area of $44\,\text{m}^2$. Find the value of x.



4

5 Solve $5x + 3 = \sqrt{3x+7}$.

5

2(h) Use the quadratic formula to solve $2x^2 + 11x = 5x^2 - 18$. Give answers to three significant figures.

2(h)

3(e) Solve $(2x + 8)^2 = 100$.

3(e)

3(f) Solve $6x^2 + 6 = 12x$.

3(f)

(i) Use $y = \sqrt{x} \geq 0$ to solve $x + 3\sqrt{x} - 1/2 = 0$.

(i)

(ii)(a) Solve $x + 10\sqrt{x + 2} - 22 = 0$.

(ii)(a)

(ii)(b) Solve $x^2 - 4x + \sqrt{2x^2 - 8x - 3} - 9 = 0$.

(ii)(b)

2: Completing the Square 11

Completing the square rewrites a quadratic as $a(x+b)^2+c$. The variable then appears only once, making several tasks easier.

Solving Solve $(x+4)^2-7=0$.

Solving

Sketching

The graph of $y=(x+a)^2+b$ has minimum point $(-a,b)$.

Later integration

In Further Maths, rewriting $x^2-4x+5=(x-2)^2+1$ helps when integrating

$$\int \frac{1}{x^2-4x+5}\,dx$$

Completing the Square Recap 12

Expand $(x+9)^2=x^2+18x+81$ and $(x-5)^2=x^2-10x+25$.

The number inside the bracket is half the coefficient of x.

1 Complete the square: x^2+12x.

1

2 Complete the square: x^2+8x.

2

3 Complete the square: x^2-2x.

3

4 Complete the square: x^2-6x+7.

4

$$ax^2+bx+c=a\left(x+\frac{b}{2a}\right)^2+c-\frac{b^2}{4a}\quad(a\neq 0)$$

Think through the steps rather than relying only on memorising this formula.

Completing the Square

Express $2x^2 + 12x + 7$ in completed-square form.

1. Factor out 2

2. Complete the square

3. Expand

4. Simplify

Express $5 - 3x^2 + 6x$ in completed-square form.

5. Factor out -3

6. Complete the square

7. Expand

8. Simplify

Test Your Understanding

1 Express $3x^2 - 18x + 4$ as $a(x + b)^2 + c$.

1

2 Express $20x - 5x^2 + 3$ as $a - b(x + c)^2$.

2

Solving by Completing the Square 15

Solve $3x^2 - 18x + 4 = 0$ by completing the square.

1. Divide both sides by 3

2. Complete the square and solve

An equation allows division of both sides by the same nonzero number. An expression must retain its value: factor out 3 instead of simply dividing it away.

Exercise 2C/2D Pure Y1 Page 23-24 16

Complete the square. In each answer, expand the brackets to check that you recover the original expression.

1(a) $x^2 + 4x =$ 1(a)

1(b) $x^2 - 6x =$ 1(b)

1(c) $x^2 - 16x =$ 1(c)

1(d) $x^2 + x =$ 1(d)

1(e) $x^2 - 14 =$ 1(e)

2(a) $2x^2 + 16x =$ 2(a)

2(b) $3x^2 - 24x =$ 2(b)

2(c) $5x^2 + 20x =$ 2(c)

2(d) $2x^2 - 5x =$ 2(d)

2(e) $8x - 2x^2 =$ 2(e)

Exercise 2C/2D Pure Y1 Page 23-24

4 Given $x^2 + 3x + 6 = (x + a)^2 + b$, find a, b.

4

5 Write $2 + 0.8x - 0.04x^2$ as $A - B(x + C)^2$.

5

Proof Complete the square to show that $x^2 + 2bx + c = 0$ has solutions $x = -b \pm \sqrt{b^2 - c}$.

Proof

Proving the Quadratic Formula

Given $ax^2 + bx + c = 0$, with $a \neq 0$, prove the quadratic formula by completing the square.

Proof

A function gives a rule that assigns one output to each permitted input. It is a formal version of a number machine.

input x $\longrightarrow$ f $\longrightarrow$ output $f(x)$

Example The function doubles its input. Write its rule.

Example

For $f(x) = x^2$, example inputs and outputs are:

x	-1	0	1.7	2	3.1
$f(x)$	1	0	2.89	4	9.61

Domain and range What do these terms mean?

Domain and range

Roots What is a root (or zero) of a function?

Roots

$x \in \mathbb{R}$ reads: x is a member of the set of real numbers.

Examples 21

Let $f(x) = x^2 - 3x$ and $g(x) = x + 5$, with $x \in \mathbb{R}$.

(a) Find $f(-4)$.

(a)

(b) Find x for which $f(x) = g(x)$.

(b)

(c) Find the roots of f.

(c)

(d) Find the root of g.

(d)

Examples 22

Find the minimum value of $f(x) = x^2 - 6x + 2$, and the value of x where it occurs.

Trying inputs suggests an answer:

x	1	2	3	4
$f(x)$	-3	-6	-7	-6

But a few examples do not prove a minimum.

Complete the square

General rule

Quickfire Questions 23

For each function, complete the square, state its minimum or maximum, and give the input where it occurs.

1

$f(x) = x^2 + 4x + 9$

1

2

$f(x) = x^2 - 10x + 21$

2

3

$f(x) = 10 - x^2$

3

4

$f(x) = 8 - x^2 + 6x$

4

Test Your Understanding 24

1 Find the minimum of $2x^2 + 12x - 5$, and where it occurs.

1

2 Find the roots of $2x^2 + 3x + 1$.

2

3 Find the real roots of $x^4 - x^2 - 6$.

3

2 Given $f(x) = x^2 - 2x$, find a when $f(a) = 8$.

2

6 Complete the square for $f(x) = x^2 - 2x + 2$. Explain why $f(x) > 0$ for all real x.

6

8 Find the roots of $f(x) = 3^{2x} - 28 \cdot 3^x + 27$.

8

4: Quadratic Graphs 26

Sketching a quadratic

Find and label the key features. A sketch does not require a uniform numerical scale.

1. Roots

Solve $ax^2 + bx + c = 0$ to find any x-intercepts.

2. Vertical intercept

Put $x = 0$: the point is $(0, c)$.

3. Shape

If $a > 0$, the parabola opens upwards. If $a < 0$, it opens downwards.

4. Turning point

Complete the square: $y = a(x - h)^2 + k$ has turning point (h, k).

The axis of symmetry is $x = h$. Draw a smooth curve through the labelled points.

Example 27

Sketch $y = x^2 + 3x - 4$, showing the intercepts and turning point.

1. Find key points

2. Sketch

Example 28

Sketch $y = 4x - 2x^2 - 3$.

1. Key features

2. Sketch

Test Your Understanding 29

Sketch each quadratic, labelling the intercepts and turning point.

$y = x^2 + 4$

a

$y = x^2 - 7x + 10$

b

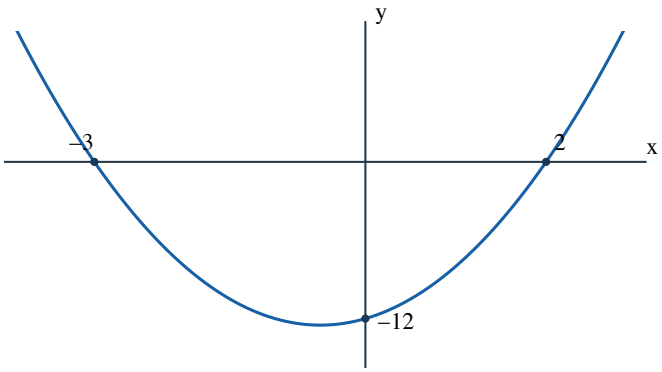
$y = 5x + 3 - 2x^2$

c

$y = x^2 + 4x + 11$

d

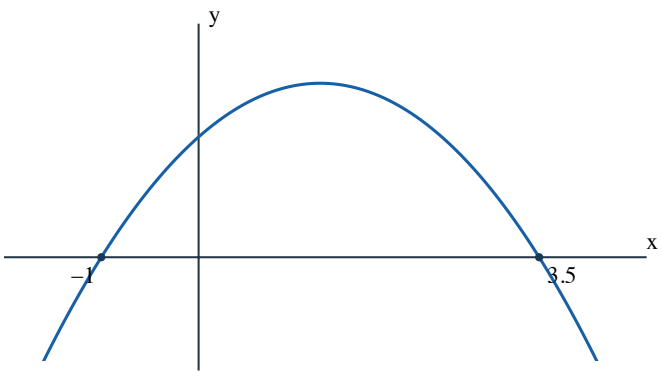
Determining the Equation using a Graph 30



Question Find the equation in the form $y = ax^2 + bx + c$.

Question

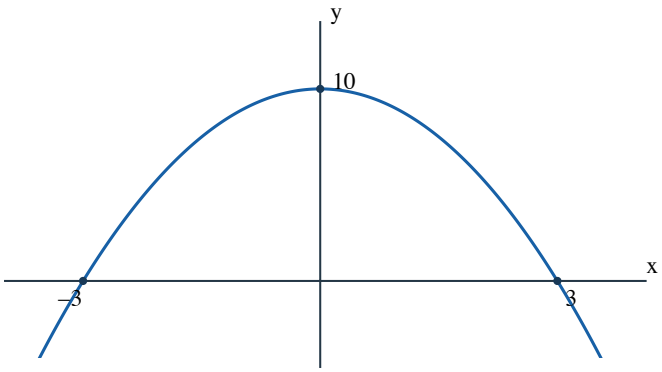
Further Example 31



Question Find one possible equation with integer coefficients.

Question

Test Your Understanding 32

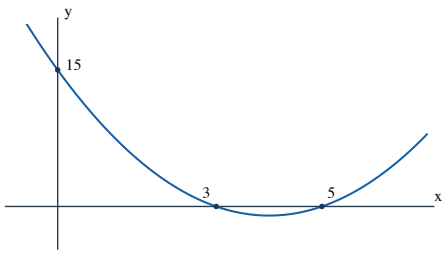


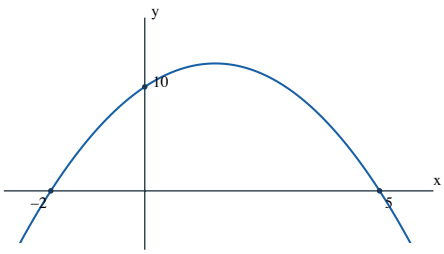
Question Determine an equation of this graph.

Question

Exercise 2F Pure Y1 Page 30 33

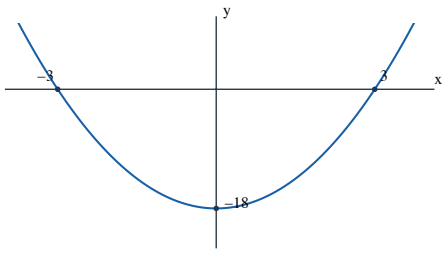
Find the equation $y = ax^2 + bx + c$ of each graph.

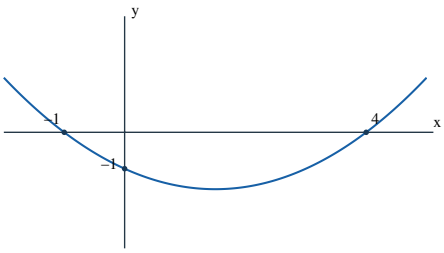




a

b





c

d

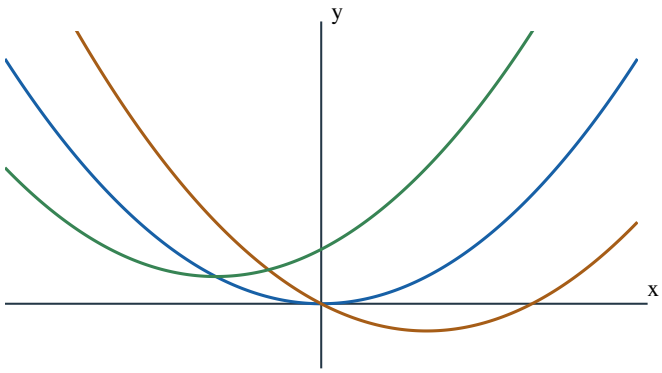
Exercise 2F Pure Y1 Page 30 34

3 The graph $y = ax^2 + bx + c$ has minimum $(5, -3)$ and passes through $(4, 0)$. Find a, b, c.

3

Extension · MAT 2003 1H

Into how many regions do $y = x^2$, $y = x^2 - 2x$, $y = x^2 + 2x + 2$ divide the plane?



Extension

Starter 35

How many distinct real solutions does each equation have?

1

$x^2 - 12x + 36 = 0$

1

2

$x^2 + x + 3 = 0$

2

3

$x^2 - 2x - 1 = 0$

3

5: The Discriminant 36

$ax^2 + bx + c = 0, \quad a \neq 0 \qquad x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

The discriminant is $D = b^2 - 4ac$.

No real roots What condition is needed?

No real roots

Equal roots What condition is needed?

Equal roots

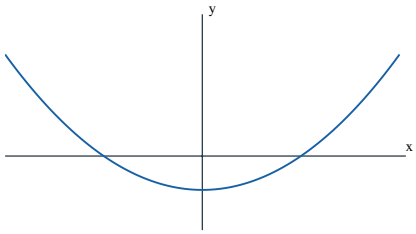
Two distinct real roots What condition is needed?

Two distinct real roots

Roots are the values of x that make the quadratic equal to zero.

Two distinct real roots

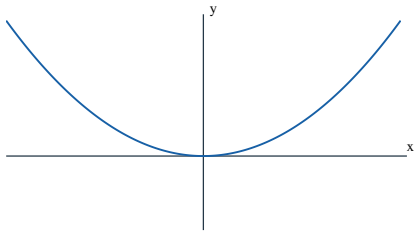
$D > 0$



The graph crosses the horizontal axis twice.

Equal roots

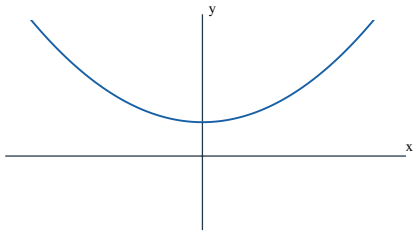
$D = 0$



The graph touches the horizontal axis at its turning point.

No real roots

$D < 0$



The graph does not meet the horizontal axis.

Why say "equal roots"?

$x^2 - 12x + 36 = (x - 6)^2$

The root 6 occurs twice as a factor. There is one distinct root, with multiplicity 2.

Fundamental theorem of algebra

Every polynomial of degree $n \geq 1$ has exactly n complex roots when counted with multiplicity.

The degree is the highest power with a nonzero coefficient. A quadratic has degree 2.

Thus a quadratic always has two complex roots counting multiplicity, even when it has no real roots.

An example of factorisation over the reals

$$x^4 + a^4 = (x^2 + \sqrt{2}ax + a^2) \times (x^2 - \sqrt{2}ax + a^2)$$

Expanding gives $(x^2 + a^2)^2 - 2a^2x^2 = x^4 + a^4$.

Every real polynomial can be factored into real linear and quadratic factors.

Quickfire Questions 39

Find the discriminant and state the number of distinct real roots.

1

$x^2 + 3x + 4 = 0$

1

2

$x^2 - 4x + 1 = 0$

2

3

$x^2 - 4x + 4 = 0$

3

4

$2x^2 - 6x - 3 = 0$

4

5

$x - 4 - 3x^2 = 0$

5

6

$1 - x^2 = 0$

6

Problems involving the discriminant 40

The equation $x^2 + 2px + (3p + 4) = 0$ has equal roots, where $p > 0$. Find p, then solve the equation.

1. Set the discriminant to zero

2. Find p

3. Find the repeated root

Exercise 2G Pure Y1 Page 32 41

1 Find the values of k for which $x^2 + 6x + k = 0$ has two distinct real roots.

1

2 Find t if $2x^2 - 3x + t = 0$ has a repeated root.

2

Exercise 2G Pure Y1 Page 32 42

1 The equation $sx^2 + 8x + s = 0$ has equal roots, with $s > 0$. Find s.

1

2 Find the values of k for which $3x^2 - 4x + k = 0$ has no real roots.

2

Exercise 2G Pure Y1 Page 32 43

6 $g(x) = x^2 + 3px + (14p - 3)$ has equal roots, and p is an integer. Find p and the root.

6

7 Show that $2x^2 + (k + 4)x + k = 0$ has two distinct real roots for every real k.

7

Test Your Understanding 44

1 Find k when $x^2 + 5kx + (10k + 5) = 0$ has equal roots.

1

2 Find k when $x^2 + 6x + k = 0$ has two distinct real roots.

2

Exercise 2G Pure Y1 Page 32

1. MAT 2011 1B A rectangle has perimeter P and area A. Which must hold?

$P^3 > A$; $A^2 > 2P + 1$; $P^2 \geq 16A$; $PA > A + P$

1. MAT 2011 1B

2. MAT 2006 1B How many real roots does $(2 + x - x^2)^2 = 16$ have?

2. MAT 2006 1B

3. MAT 2009 1C For which real c does $x^4 = (x - c)^2$ have four real roots, counting repetitions?

3. MAT 2009 1C

Modelling 46

A spear is thrown from a tower over level ground. Its height in metres after t seconds is modelled by

$h(t) = 12.25 + 14.7t - 4.9t^2, \quad t \geq 0$

A mathematical model describes a situation using simplifying assumptions.

(a) Interpret 12.25.

(a)

(b) When does the spear hit the ground?

(b)

(c) Write $h(t)$ as $A - B(t - C)^2$.

(c)

(d) Find the maximum height and when it is reached.

(d)

Exercise 2H Pure Y1 Page 34 — 47

A car model predicts fuel economy y mpg at speed x mph:

$y = -0.01x^2 + 0.975x + 16, \quad x > 0$

(a) Find the two speeds giving 32.5 mpg.

(a)

(b) Rewrite as $A - B(x - C)^2$.

(b)

(c) At what speed is fuel economy greatest?

(c)

(d) Calculate the prediction at 120 mph and comment on validity.

(d)

Exercise 2H Pure Y1 Page 34 — 48

Grain yield g, in tonnes per hectare, is modelled using fertiliser f, in kilograms per hectare:

$g = 6 + 0.03f - 0.00006f^2$

(a) Find the yield without fertiliser.

(a)

(b) A farmer currently uses 20 kg per hectare. Find the smallest additional amount needed to increase yield by 1 tonne per hectare.

(b)

Exercise 2H Pure Y1 Page 34 49

A stadium has 25,000 seats. At £30 per ticket, 10,000 tickets sell. Each £1 reduction sells 1,000 more tickets. Model sales by $t = M - 1000p$, where £p is the ticket price.

(a) Find M.

(a)

(b) Revenue is $r = p(M - 1000p)$. Write it as $A - B(p - C)^2$.

(b)

(c) Find the price that maximises revenue.

(c)

Would you like $1,000,000 for finding roots? 50

Roots are inputs that give an output of zero. Finding roots can also lead to major research problems.

For real $s > 1$, the Riemann zeta function can be defined by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

$$\zeta(2) = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots$$

The function extends to complex inputs through a more advanced definition.

The Riemann hypothesis

It predicts that every nontrivial zero of the extended zeta function has real part $1/2$.

This is a Millennium Prize Problem with a US$1 million prize for a qualifying solution. It connects zeros of a function with the distribution of prime numbers.

Read the Clay Mathematics Institute explanation